Designing Clinical Research

cost of an ancillary study is much less than the cost of conducting the same study independently. Some large studies may have their own mechanisms for funding ancillary studies, especially if the research question is important and considered relevant by the funding agency.

The **disadvantages** of ancillary studies are few. In some cases there may be practical problems in obtaining official permission to perform the study, training those who will make the measurements, and obtaining separate informed consent from participants. Because the ancillary study investigator may not have designed or conducted the main study, it may also be difficult to obtain access to the full database for analysis. These issues, including a clear understanding of authorship of scientific papers that result from the ancillary study and the rules governing their preparation and submission, need to be clarified before starting the study.

■ SYSTEMATIC REVIEWS

Systematic reviews identify completed studies that address a research question, and evaluate the results of these studies to arrive at conclusions about a body of research. In contrast to other approaches to reviewing the literature, systematic reviews use a well-defined and uniform approach to identify all relevant studies, display the results of eligible studies, and, when appropriate, calculate a summary estimate of the overall results. The statistical aspects of a systematic review (calculating summary effect estimates and variance, statistical tests of heterogeneity, and statistical estimates of publication bias) are called **meta-analysis.**

A systematic review can be a good opportunity for a new investigator. Although it takes a surprising amount of time and effort, a systematic review generally does not require substantial financial or other resources. Completing a good systematic review requires that the investigator become intimately familiar with the literature regarding the research question. For new investigators, this detailed knowledge of published studies is invaluable. Publication of a good systematic review can also establish a new investigator as an "expert" on the research question. Moreover, the findings, with power enhanced by the larger sample size available in the combined studies and peculiarities of individual study findings revealed by comparison with the others, often represent an important scientific contribution. Systematic review findings can be particularly useful for developing practice guidelines.

The elements of a good systematic review are listed in Table 13.3. Just as for

■ TABLE 13.3
Elements of a Good Systematic Review

1. Clear research question
2. Comprehensive and unbiased identification of completed studies
3. Definition of inclusion and exclusion criteria
4. Uniform and unbiased abstraction of the characteristics and findings of each study
5. Clear and uniform presentation of data from individual studies
6. Calculation of a summary estimate of effect and confidence interval based on the findings of all eligible studies when appropriate
7. Assessment of the heterogeneity of the findings of the individual studies
8. Assessment of potential publication bias
9. Subgroup and sensitivity analyses

other studies, the methods for completing each of these steps should be described in a written protocol before the systematic review begins.

The Research Question

As with any research, a good systematic review has a well-formulated, clear research question that meets the usual FINER criteria (feasible, interesting, novel, ethical, and relevant; see Chapter 2). Feasibility depends largely on the prior existence of a set of studies of the question. The research question should describe the disease or condition of interest, the population and setting, the intervention and comparison treatment (for trials), and the outcomes of interest. For example, "*Among persons admitted to an intensive care unit unstable angina, does treatment with aspirin plus intravenous heparin reduce the risk of myocardial infarction and death during the hospitalization more than treatment with aspirin alone* (17)?"

Identifying Completed Studies

Systematic reviews are based on a comprehensive and unbiased search for completed studies. The search should follow a well-defined strategy established before the results of the individual studies are known. The process of identifying studies for potential inclusion in the review and the sources for finding such articles should be explicitly documented before the study. Searches should not be limited to MEDLINE, which includes only about half of all published English-language clinical research studies and often does not list non-English-language references. Depending on the research question, other electronic databases such as AIDSLINE, CANCERLIT, and EMBASE can be included, as well as manual review of the bibliography of relevant published studies, previous reviews, evaluation of the Cochran Collaboration database, and consultation with experts.

Criteria for Including and Excluding Studies

The protocol for a systematic review should provide a good rationale for including and excluding studies, and these criteria should be established a priori. Criteria for including or excluding studies from meta-analyses typically designate the period during which studies were published, the population that is acceptable for study, the disease or condition of interest, the intervention to be studied, whether blinding is required (for trials), acceptable control groups, required outcomes, maximal acceptable loss to follow-up, and minimal acceptable length of follow-up. Once these criteria are established, each potentially eligible study should be reviewed for eligibility independently by two or more investigators, with disagreements resolved by another reviewer or by consensus. When determining eligibility, it may be best to blind reviewers to the date, journal, authors, and results of trials.

Published systematic reviews should list studies that were considered for inclusion and the specific reason for excluding a study. For example, if 30 potentially eligible trials are identified, these 30 trials should be fully referenced and reasons should be given for each exclusion.

Collecting Data from Eligible Studies

Data should be abstracted from each study in a uniform and unbiased fashion. Generally, this is done independently by two or more abstractors using predesigned forms that include variables that define eligibility criteria, design features, the population included in the study, the number of individuals in each group, the intervention (for trials), the main outcome, secondary outcomes, and outcomes in subgroups. The data abstraction forms should include any data that will subsequently appear in tables describing the studies included in the systematic review or in tables or figures presenting the outcomes. When the two abstractors disagree, a third abstractor may settle the difference, or a consensus process may be used. The process for abstracting data from studies for the systematic review should be clearly described in the manuscript.

The published reports of some studies that might be eligible for inclusion in a systematic review may not include important information, such as design features, risk estimates, and standard deviations. Often it is difficult to tell if design features such as blinding were not implemented or were just not described in the publication. The reviewer can sometimes calculate relative risks and confidence intervals from crude data presented from randomized trials, but it is generally unacceptable to calculate risk estimates and confidence intervals based on crude data from observational studies unless there is sufficient information to adjust for potential confounders. Every effort should be made to contact the authors to retrieve important information that is not included in the published description of a study. If this information cannot be calculated or obtained, the study findings are generally excluded.

Presenting the Findings Clearly

Systematic reviews generally include three types of information. First, important characteristics of each study included in the summary review are presented in tables. These often include the study sample size, number of outcomes, length of follow-up, characteristics of the population studied, and methods used in the study. Second, the review displays the results of the individual studies (risk estimates and confidence intervals) in a figure. Finally, in the absence of significant heterogeneity (see below), the review presents summary estimates and confidence intervals based on the findings of all the included studies as well as sensitivity and subgroup analyses.

The summary effect estimates represent the main outcomes of the systematic review but should be presented in the context of all the information abstracted from the individual studies. The characteristics and findings of individual studies included in the systematic review should be displayed clearly in tables and figures so that the reader can form opinions that do not depend solely on the statistical summary estimates.

Meta-Analysis: Statistics for Systematic Reviews

Summary Effect Estimate and Confidence Interval. Once all completed studies have been identified, those that meet the inclusion and exclusion criteria have been chosen, and data have been abstracted from each study, a summary estimate (summary relative risk, summary odds ratio, etc.) and confidence interval may be calculated. The summary effect is essentially an average effect weighted by the size of each study. Methods for calculating the summary effect and confidence interval are discussed in Appendix 13.1.

Heterogeneity. Combining the results of several studies is not appropriate if the studies differ in clinically important ways, such as the intervention, outcome, controls, blinding, and so on. It is also inappropriate to combine the findings if the results of the individual studies differ widely. Even if the methods used in the studies appear to be similar, the fact that the results vary markedly suggests that something important was different in the individual studies. This variability in the findings of the individual studies is called **heterogeneity** (and the study findings are said to be **heterogeneous**); if there is little variability, the study findings are said to be **homogeneous.**

How can the investigator decide if the studies used the same methods and had similar findings? First, he can review the individual studies to determine if there are substantial differences in study design, study populations, intervention, or outcome. Then he can examine the results of the individual studies. If some trials report a substantial beneficial effect of an intervention and others report considerable harm, heterogeneity is clearly present.

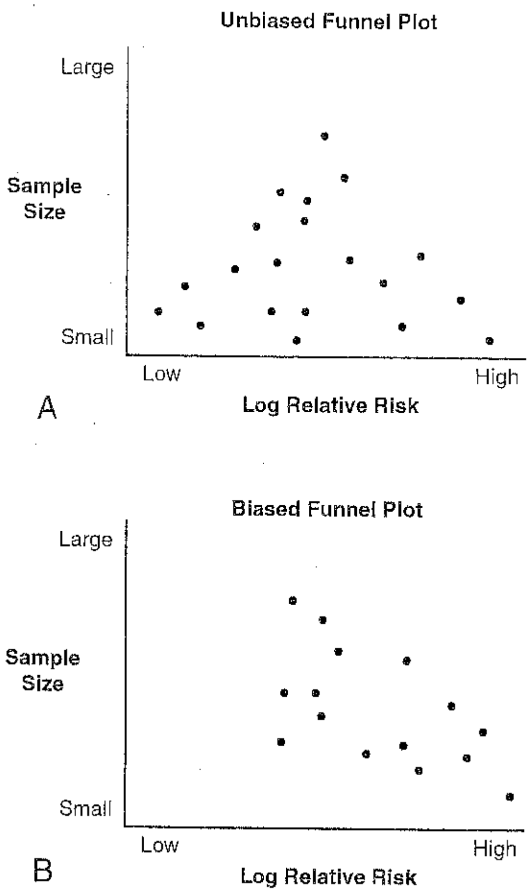
Sometimes, however, it is more difficult to decide if heterogeneity is present. For example, if one trial reports a 50% risk reduction for a specific intervention but another reports only a 30% risk reduction, is heterogeneity present? Statistical approaches (tests of homogeneity) have been developed to help answer this question (Appendix 13.1).

Assessment of Publication Bias

Publication bias occurs when published studies are not representative of all studies that have been done, usually because positive results tend to be submitted and published more often than negative results. There are two main ways to deal with publication bias. **Unpublished studies can be identified** and the results included in the summary estimate. Unpublished results may be identified by querying investigators and reviewing abstracts, meeting presentations, and doctoral theses. The results of unpublished studies can be included with those of the published trials in the overall summary estimate, or sensitivity analyses can determine if adding these unpublished results substantially changes the summary estimate determined from published results. However, including unpublished results in a systematic review is problematic for several reasons. It is often difficult to identify unpublished studies and even more difficult to abstract the required data. Frequently, inadequate information is available to determine if the study meets inclusion criteria for the systematic review or to evaluate the quality of the methods. For these reasons, unpublished data are not often included in meta-analyses.

Alternatively, the extent of potential **publication bias can be estimated** and this information used to temper the conclusions of the systematic review. Publication bias exists when unpublished studies have different findings from published studies. Unpublished studies are more likely to be small (large studies usually get published, regardless of the findings) and to have found no association between the risk factor or intervention and the outcome (markedly positive studies usually get published, even if small). If there is no publication bias, there should be no association between a study's size and findings. A strong correlation between study outcome and sample size suggests publication bias. In the absence of publication bias, a plot of study sample size (or study weight) versus outcome (e.g., log relative risk) should have a bell or funnel shape with the apex near the summary effect estimate.

The funnel plot in Fig. 13.1A suggests that there is little publication bias because small studies with both negative and positive findings were published. The plot in Fig. 13.1B, on the other hand, suggests publication bias because the distribution appears truncated in the corner that should contain small, negative studies.



■ **FIGURE 13.1**

A: Funnel plot suggestive of minimal publication bias since there are studies with a range of large and small sample sizes, and low relative risks are reported by some smaller studies. **B:** Funnel plot suggestive of publication bias since the smaller studies primarily report high relative risks.

When substantial publication bias is likely, summary estimates should not be calculated or should be interpreted cautiously. Every reported systematic review should include some discussion of potential publication bias and its effect on the summary estimates.

Subgroup and Sensitivity Analyses

Subgroup analyses may be possible using data from all or some subset of the studies included in the systematic review. For example, in a systematic review of the effect of postmenopausal estrogen therapy on endometrial cancer risk, some of the studies presented the results by duration of estrogen use. Subgroup analyses of the results of studies that provided such information demonstrated that longer duration of use was associated with higher risk for cancer (18).

Sensitivity analyses indicate how "sensitive" the findings of the meta-analysis are to certain decisions about the design of the systematic review or inclusion of certain studies. For example, if the authors decided to include studies with a slightly different design or methods in the systematic review, the findings are strengthened if the summary results are similar whether or not the questionable studies are included. Systematic reviews should generally include sensitivity analyses if any of the design decisions appear questionable or arbitrary.

Garbage In, Garbage Out

The biggest drawback to a systematic review is that it can produce a very reliable-appearing summary estimate based on the results of individual studies that are of poor quality. The process of assessing quality is complex and problematic. We favor relying on relatively strict criteria for good study design when setting the inclusion criteria. If the individual studies that are summarized in a systematic review are of poor quality, no amount of careful analysis can prevent the summary estimate from being unreliable.

■ SUMMARY

Secondary Data Analysis

1. Secondary data analysis has the **advantage** of greatly reducing the time and cost of doing research and the **disadvantage** of providing the investigator with little or no control over the data.
2. One good source of data for secondary analysis is an **existing project** at the investigator's institution; another is the large number of **databases** now available from many sources. Although **individual data are preferable**, aggregate data can also sometimes be useful for ecologic analysis.
3. Investigators may begin either by looking for research questions to fit an existing database or by looking for a database that can answer a particular research question.
4. Large community-based data sets are useful for studying the **effectiveness** and **utilization** of an intervention in the community, and for discovering **rare adverse events**.

Ancillary Studies

1. A clever ancillary study can answer a new research question with **little cost and effort.** As with secondary data analyses, the investigator **cannot control the design,** including the population and many of the variables measured, but he is able to **specify a few key additional measurements.**
2. Good opportunities for ancillary studies may be found in **cohort studies** or **clinical trials** that include either the predictor or outcome variable for the research question of interest. **Stored banks of serum, DNA, images,** and so on, provide the opportunity for cost-effective nested case-control designs.
3. Most large studies have written procedures that allow investigators (including outside scientists) to propose and carry out ancillary studies.

Systematic Reviews

1. A good systematic review, like any other study, requires a complete **written protocol** before the study begins. The protocol should include the **research question,** methods for **identifying all eligible studies,** methods for **abstracting data** from the studies, and **statistical methods.**
2. The statistical aspects of a systematic review, termed **meta-analysis,** include the **summary effect estimate and confidence interval,** tests for evaluating **heterogeneity** and **potential publication bias,** and planned **subgroup** and **sensitivity analyses.**

3. The **characteristics** and **findings** of individual studies should be displayed clearly in tables and figures so that the reader can form opinions that do not depend solely on the statistical summary estimates.
4. The biggest drawback to a systematic review is that the results can be no more reliable than the **quality** of the underlying studies upon which it is based.

EXERCISES

1. The research question is, "Do Latinos in the United States have higher rates of gallbladder disease than whites, African Americans, or Asian Americans?" What existing databases might enable you to determine race-, age- and sex-specific rates of gallbladder disease at low cost in time and money?

2. A research fellow became interested in the controversial question of whether high serum triglyceride levels increase risk for coronary heart disease. Because of the expense and difficulty of conducting a study to generate primary data, he wanted an existing database that contained the variables he needed to answer his research question but was generated for other reasons (so that it was unlikely that someone else had already studied the research question with that data set). He found an investigator at his institution who had access to the database of a large multicenter randomized controlled trial of the effects of interventions on several risk factors for coronary heart disease (the Multiple Risk Factor Intervention Trial).

Fortunately, the MRFIT investigators agreed to allow him access to the data. The necessary predictors and outcome measures were present in the database. The fellow and his advisor then requested the analyses that they wanted from the MRFIT Coordinating Center.

a. What are the advantages of this approach to study this question?
b. What are the disadvantages of this approach?

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